

Summarised Resume': Traiano Welcome

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1. Personal Details

Family Name	Welcome
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2. Professional Experience (Summary)

The following table summarizes roles I have held during 25 years of formal employment.

Item	Role	Company	Start Date	End Date	Location
1	Independent Contractor / A.I Factory Architect	n/a	July 2026	Current	Thailand
2	Senior Manager, A.I Solution Architecture	Firmus Metal International Pte. Ltd	January 2026	June 2026	Singapore
3	Artificial Intelligence Infrastructure Solution Architect	Firmus Metal International Pte. Ltd	June 2025	January 2026	Singapore
3	Senior Manager, I.T Infrastructure Transformation	Accenture	December 2021	June 2025	Singapore
4	Senior Cloud Operations Engineer	Standard Chartered Bank	November 2020	December 2021	Singapore
5	Senior Site Reliability Engineer	Prudential Software Services/Prudential Asia	October 2018	November 2020	Singapore
6	Systems Engineer	Teralytics	October 2017	October 2018	Singapore
7	Platform Architect	noon ecommerce (under Emaar Group)	November 2016	August 2017	Dubai
8	Senior Systems Engineer (Infrastructure Services Team)	MH Alshaya Co. WLL	June 2014	June 2016	Kuwait
9	Operations Engineer (EC2 Platform)	Amazon Web Services	May 2013	March 2014	South Africa
10	Cloud Support Engineer	Amazon Web Services	September 2012	May 2013	South Africa
11	UNIX Systems Engineer (Infrastructure Services)	MTN Business Solutions Pty. Ltd	January 2010	September 2012	South Africa
12	Network Monitoring Specialist	Tertiary Education and Research Network of South Africa (TENET)	March 2008	December 2009	South Africa

13	Technical Business Developer	Lightedge Technologies Pty. Ltd	Jan 2008	March 2008	South Africa
14	System Administrator	Lightedge Technologies Pty. Ltd	March 2007	Jan 2008	South Africa
15	System Administrator (Infrastructure Services Team)	Internet Solutions Pty. Ltd	January 2005	February 2007	South Africa
16	Network Engineer	Qualica Technologies Pty. Ltd	May 2001	December 2004	South Africa

This section provides a listing of a few of the key projects I have implemented or taken the lead on during my employment. Note: These projects are often run concurrently.

Item	Project	Description	Company
1	Implemented Multitenant Architecture for bare-metal GPUaaS sub-letting	Design a model for dividing a customer tenant into further sub-tenants for GPUaaS “subletting”, including network, security, access, RBAC and storage. Assist customer in implementing the model.	Firmus
2	GPUaaS Security and Risk Model	Develop the business case, requirements security model for H200 GPUs provided over the public internet on bare metal and virtualized platforms. Manage the implementation of security layer for HPC clusters.	Firmus Metal International/SMC
3	Public Neocloud Platform for GPUaaS	Design and manage the implementation of a multi-tenant public cloud platform for delivery of H200 GPUaaS. Develop the market strategy and customer onboarding process. Develop the long term evolution and roadmap of the NCP Cloud Platform. Develop the operational model and vendor management model. Manage the relationship with platform customers during the architecture process. Onboard and evaluate the first commercial customers on the platform.	Firmus Metal International/SMC

4	Nvidia-based HPC platform for GenAI Factory (250-node GB300 NV72 cluster)	Develop the high level architecture for 250-node GB300 HPC cluster and manage the build and validation of the platform to customer handover. Develop the operational model. Manage the relationship with the customer during the architecture process.	Firmus Metal International/SMC
5	HPC Storage Cluster for high parallelism (VAST Data Cluster)	Develop requirements, business case, solution architecture and manage build for VAST Storage cluster for GenAI HPC cluster platforms in Australian datacenters.	Firmus Metal International/SMC
6	WEKA PFS cluster for shared HPC storage Platform	Develop requirements, business case, solution architecture and manage build for new WEKA Storage clusters for GenAI HPC cluster platforms in Singapore datacenters.	Firmus Metal International/SMC
7	Nvidia-based HPC platform for Generative A.I Factory (72-node H200)	Develop the high level architecture for 72-node H200 HPC cluster and manage the build and validation of the platform to customer handover. Develop the security model. Manage the relationship with the customer during the architecture process.	Firmus Metal International/SMC
8	NETS/PayNow/GIRO infrastructure analysis and development roadmap	Produce a complete analysis of the Singapore NETS/PayNow payments infrastructure to root cause payments failures identified by the MAS (Monetary Authority of Singapore). Produce recommendations on remediating systemic platform issues and future development roadmap of the payments platform.	Accenture Customer (Monetary Authority of Singapore)
9	Accenture A.I Refinery (“Generative A.I Factory”)	Architect and manage the build of components of the Accenture A.I Refinery on Azure Cloud	Accenture
10	Azure Cloud Landing Zone and Security Framework	Infrastructure as Code and deployment automation of Azure CAF landing zones and security framework implementation in code	Accenture Customer (Large betting firm in Asia)

11	Azure Cloud Governance and Cloud Operating Model Optimisation	Analyse and optimise the enterprise Cloud Governance and Cloud Operating Model for the Business on Azure Cloud	Accenture (Large Insurance customer in Asia)
12	Trading System Transaction Load Testing Solution	Implement a large-scale Cloud based event driven platform for benchmarking Trading systems for customers specialising in Trading platforms.	Accenture (Crypto Currency Exchange Customer)
13	Azure Cloud Operational Acceptance Testing Framework and Process	Implement a Process and Tools to put new applications through an operational acceptance testing process for deployment to SCBs production cloud environment	Standard Chartered Bank
14	FinOps Analytics Solution for Azure Cloud	Integrate ApptIO and Cloudability SaaS Analytics solution for FinOps implementation	Standard Chartered Bank
15	Hybrid Cloud Foundations	Foundational Infrastructure as Code Implementation for all Prudential Business Units in Asia (Terraform+BASH+Ansible+PowerShell)	Prudential Asia/PSS
16	Kubernetes Backup and Disaster Recovery Solution	Automated backup solution for providing data backups for applications running on Kubernetes in Azure (based on Velero)	Prudential Asia/PSS
17	Conversational Chatbot implementation for Azure	Infrastructure as code implementation of Conversational AI Chatbot module for use across Prudential	Prudential Asia/PSS
18	SQL MI modular deployment automation	Terraform based module for deployment of SQL Managed Instances to Azure Cloud and Jenkins pipelines to execute the deployment and configuration	Prudential Asia/PSS
19	Azure App Service deployment automation	Development of Infrastructure As Code implementation of Azure Application Service in modular format and Jenkins pipelines for deployment to Cloud	Prudential Asia/PSS
20	Azure Data Lake Implementation	Implement a collection of data processing resources in Terraform for a Data Lake prototype	Prudential Asia/PSS

21	CI/CD Deployment for Cloud Infrastructure	Architect CI/CD solution (Jenkins) and pipelines for deploying Cloud infrastructure as Code (Terraform, Ansible, Bash, Powershell, Python)	Prudential Asia/PSS (2020/2019)
22	Azure Kubernetes Implementation	Architect, Develop and Implement an AKS Infrastructure deployment model for all Prudential Business Units	Prudential Asia/PSS (2020/2019)
23	BYOK implementation for MS SQL PaaS and Managed instances	Implement a BYOK deployment automation for Azure MS SQL PaaS and Managed Instances using Azure Key Vault	Prudential Asia/PSS (2020/2019)
24	Foundational Terraform Modules for Infrastructure as Code	Write a range of Terraform modules for business users to use in deploying applications to Prudential's Cloud (E.g storage, Keyvaults, AKS, Azure Shares, Azure Postgresql, Azure CosmosDB)	Prudential Asia/PSS (2020/2019)
25	CMDB for Infrastructure as Code Automation	Design schema and implement a CMDB to support CD pipelines for deploying infrastructure code	Prudential Asia/PSS (2020/2019)
26	Jenkins CI/CD server implementation on RedHat OpenShift	Implementation of container based Jenkins CI/CD server using OpenShift with containerised workers as well as server based workers. Implementation of Docker images for worker deployer nodes.	Prudential Asia/PSS (2020/2019)
27	Hardware based Data Analytics cluster	Hardware based Data Analytics cluster based on apache mesos, docker, linux and HDFS, supporting kafka and spark streaming. Supported internal development needs of the Data Science team	Teralytics (2018/2017)
28	VM based Data Analytics cluster	Virtual Machine based Data Analytics cluster based on apache mesos, docker, linux and HDFS, supporting kafka and spark streaming. Supported internal development needs of M1 Internet Service provider's analytics projects	Teralytics (2018/2017)

29	DC/OS implementation design proposal	Architecture and Proposal of a replacement of Apache Mesos stack with Mesosphere's DC/OS product for tighter integration of components and smoother operation	Teralytics (2018/2017)
30	GitLab Design Proposal for a new CI/CD system to replace Jenkins	Design and demonstration of GitLab as an alternative CI/CD solution to Jenkins for Infrastructure Deployment	Teralytics (2018/2017)
31	DC/OS Containerised Microservices Platform Implementation	Design and Implementation of DC/OS using Terraform on the AWS Cloud to support Microservices development for ecommerce (Vendor partner Mesosphere GMBH)	noon.com (2016/2015)
32	Centralised Logging Solution	Centralised logging solution for I.T infrastructure based on Syslog-NG and Splunk (Vendor Partner Splunk and BalaBit)	noon.com (2016/2015)
33	Hadoop Cluster Deployment for Enterprise Data Hub (PoC)	Proving of Cloudera Hadoop and MAPR, and Building of Production Cluster for Enterprise Data Hub	MH Alshaya (2015/2014)
34	Centralised Authentication / SSO for Linux and UNIX	Testing, Design, Rollout Planning, Rollout and Maintenance of Centralised Authentication Solutions based on Centrify and Redhat IDM/FreeIPA	MH Alshaya (2015/2014)
35	Centralised Logging and User activity Auditing	Design, Requirements analysis and rollout planning of Syslog, Elastic Search + Kibana	MH Alshaya (2015/2014)
36	VM Deployment Automation	Template Build Automation and deployment automation via scripted methods	MH Alshaya (2015/2014)
37	Enterprise Password Management Solution	Password Vault and Sharing Solution implementation and administration for corporate system passwords. Based on Manage Engine's Password Manager	MH Alshaya (2015/2014)

38	MS SQL Cluster Implementation	Requirements Analysis, Design and deployment of a High Availability MSSQL Cluster Platform on Hyper-V, for consolidation of Standalone MS SQL Servers.	MH Alshaya (2015/2014)
39	High Availability RADIUS Cluster for VOIP and ADSL networks	Design, PoC and Implementation of a Highly Available RADIUS cluster based on RADIATOR, and FreeBSD	MTN
40	Distributed Monitoring System for Servers and Applications	Design, Implementation and administration of a distributed, scalable monitoring platform based on Nagios, Netcool and Net-SNMP.	MTN
41	Network Monitoring and Management Solution for SONET Fibre Optic Ring and ADSL	Design, Implementation and operation of a distributed, flexible monitoring and reporting platform for ISP ADSL and Fibre networks based on Cacti, Nagios, Net-SNMP.	TENET
42	CMDB – Configuration Management database for network and server management	Design, Implementation and Operation of a Federated CMDB for keeping rack of ISP system configuration	TENET
43	Highly Available Distributed Bulk Email processing cluster based on exim, irnport/brightmail, postfix, mysql, spamassassin.	Design, Implementation and Administration of a geographically distributed, highly available, high performance bulk SMTP mail relaying cluster, with advanced spam and antivirus filtering.	Internet Solutions
44	Redundant SMTP Internal Email Relays	Redundant pair of email relays to accept mail from the internet and relay to microsoft exchange with Virus scanning and antispam. Based on Exim. FreeBSD, Postgres, Postfix, spam assassin	Internet Solutions

Independent AI Infrastructure & AI Factory Architect

As an independent consultant, I specialise in the architecture, design and technical validation of next-generation AI Factories, GPU cloud platforms and high-performance AI infrastructure.

I advise organisations on the planning, deployment and optimisation of NVIDIA-based AI compute environments, including bare-metal GPU clusters, Kubernetes platforms, high-performance storage, networking and distributed AI training infrastructure.

My work spans technical strategy, solution architecture, infrastructure validation, automation and operational readiness, helping clients design scalable, production-ready AI platforms while bridging the gap between executive business objectives and deep engineering implementation.

Senior Manager Solutions Architecture - Firmus

- Manage a global team of senior solutions architects across North America, Singapore and Australia in the specialisation of Generative A.I HPC Infrastructure.
- Lead the architecture of HPC Infrastructure for Generative A.I Factories and support the successful build and delivery of A.I HPC Platforms to Government and Global Hyperscalers.
- Develop the requirements, solution architecture and implementation plan for GB300 NVL72 clusters for customers like Meta, Nvidia, Anthropic and ByteDance.
- Support the CTO in defining the technology roadmap and evolution of the A.I factory concept
- Support the CEO in developing the economic model for Neocloud GPUaaS platforms, ensuring maximum utilisation of GPUs to achieve profitability
- Coordinate engineering, operations, sales teams and suppliers to achieve the delivery of GPU platforms to meet customer requirements.
- Ensure existing GPUaaS platforms remain compliant with developing customer requirements

A.I Infrastructure Solution Architect – Firmus/SMC

- Design and deploy scalable, high-performance AI clusters tailored to customer needs.

- Act as the technical link between customers, engineering, and product teams to support global rollout of the Firmus AI Factory and tailored A.I HPC cluster solutions.
- Lead the development and operation of a Neocloud platform for providing GPUaaS via a public cloud model using H200 GPUs.

Key Responsibilities:

- Architecting GPU-accelerated infrastructure, guiding customers through AI solution design and deployment, and engaging with both technical teams and C-level stakeholders.
- Lead and coordinate the Solution Architecture process for internal projects with vendors, partners and internal teams
- Optimise for performance and cost, advise on best practices, and lead infrastructure projects from concept to execution.
- Bring deep expertise in modern AI accelerators (e.g., NVIDIA, AMD, Intel), InfiniBand, RoCE, DPUs, and cutting-edge AI infra technologies
- Empower customers to scale AI workloads efficiently and effectively on the best of breed A.I HPC infrastructure.
- Understand the impact of changes in GPU supplier roadmap on future technology plans.

Specific Technologies used

Technology Layer	Item 1	Item 2	Item 3	Item 4
A.I / LLM Platforms	LLaMA-3 / LLaMA-4	vLLM / SGLang	NVIDIA NIM / Triton	PyTorch Distributed (DDP / Inference)
GPU Compute	NVIDIA H200 / GB300	NVIDIA DGX / HGX	NVLink / NVSwitch	Multi-GPU / Model Parallelism (FairScale)
AI Networking (Fabric)	NVIDIA ConnectX-7 / ConnectX-8	InfiniBand (Quantum)	RoCE / Spectrum-X	BlueField DPU / SuperNIC
AI Storage	WEKA Parallel File System	VAST Data Platform	Dell PowerScale (OneFS)	NVMe / NVMe-oF
Data & AI Pipelines	Kafka / Event Streaming	Object Storage (S3-compatible)	Distributed Data Loading Pipelines	Checkpointing / Dataset Staging
Observability & Telemetry	DCGM (GPU telemetry)	Prometheus / Grafana	OpenTelemetry	Cluster-level logging (Loki / ELK)
Automation & IaC	Terraform	Ansible	Python	Bash
Containerisation & Orchestration	Kubernetes	Rafay Platform	Docker / Container Runtime	Helm
Control Plane & API Layer	OpenAPI-based Control Plane	Kong / API Gateway	NVIDIA BMM / NiCo (Carbide)	Event Streaming (Kafka)
Security & Governance	Palo Alto Cortex XDR	IAM / RBAC Models	Network Segmentation (Zero Trust)	Audit / Compliance Frameworks
Cloud / Platform Integration	Hybrid Cloud (Azure / AWS integration)	Bare Metal GPUaaS	KVM-based Virtualisation	Multi-tenant Platform Design
Performance & Benchmarking	GenAI-Perf	MLPerf Inference	perf_analyzer	Custom Benchmarking Scripts
DevOps & CI/CD	Git-based pipelines	Infrastructure CI/CD	Canary / Rolling Deployments	Config Management

Senior Transformation and Cloud Implementation Manager - Accenture

- Led advanced design and implementation initiatives for large-scale AI and HPC infrastructure, specializing in NVIDIA GPU architectures (H200, GB300, DGX), InfiniBand and Spectrum-X networking, and high-performance storage platforms such as WEKA, VAST, and Dell PowerScale.

- Developed deep expertise in multi-tenant GPU-as-a-Service architectures, composable clusters, and secure cloud automation integrating Azure Entra, Rafay, HubSpot, Palo Alto cortex XDR and Monetize360.
- Directed MLPerf benchmarking and performance optimization for LLaMA-4 and other LLM workloads, using GenAI-Perf, perf_analyzer, and distributed PyTorch on multi-GPU nodes.
- Produced RFP-ready technical documentation, solution blueprints, and operational runbooks aligning with NVIDIA DGX Cloud and Meta RFP standards.
- Demonstrated strong cross-domain capability spanning AI infrastructure design, DevOps automation, benchmarking, security policy orchestration, and product-grade HPC system architecture, combining technical depth with solution architecture, vendor coordination, and enterprise-grade delivery excellence.

My key achievements during this time have been:

- Design and Implementation of an NCP Cloud (Firmus' GPUCloud) to deliver GPUaaS to the Public within a 3-month period.
- Development of Requirements and Specifications for DGX Cloud Lepton NCP Provider in collaboration with Nvidia.
- Development of Requirements and Specifications for DGX Cloud Lepton NCP Provider in collaboration with Meta.
- RFP solutioning of Infiniband, Gigabit Ethernet storage and access networks and VAST/WEKA storage solutions for a large Singaporean State I.T provider.
- Solutioning of an Nvidia HPC cluster for Nanyang Technical University, as well as inter-datacenter connectivity solutions within Singapore.
- Development of WEKA storage solutions for Singaporean organisations AISG and ACRC.

My core technology stack during this time has centered upon the following:

- Inference Engines: Llama3, Llama4, vLLM hosting automation and vLLM, SGLang
- A.I Networking: Nvidia Connectx-7, Bluefield DPU/ SuperNIC, Nvidia Quantum Infiniband Switches
- A.I Storage: WEKA and VAST Storage Systems, Dell PowerScale
- GPU Compute: Nvidia H200, GB300, L40S
- NCP/Private Cloud Orchestration: Rafay, CloudSigma, KVM, FortiGate 400F, Monetise360, Hubspot

- GPU HPC Platforms: NVIDIA DGX/HGX. Nvidia DGX Lepton. DPU. WEKA Storage.
- A.I Orchestration. AIOps/MLOps: Hybrid Cloud Based AI/ML-Ops and Machine Learning Platforms for Data Analytics
- GPUaaS Platform Security Frameworks and Implementation based on Palo Alto Cortex XDR

As a senior manager for Cloud Migration, IT Transformation and Cloud Solution Implementation under Accenture my primary responsibility was to provide Cloud and A.I/M.L transformation architecture, planning and implementation for Accenture's enterprise, government and MNC customers.

My key achievements during this time have been:

- **Technical Management of a Generative A.I Product Development Team:** Managed a dynamic team of eight engineers producing rapid prototypes of Generative A.I solutions for government and banking clients. The team was multi-skilled ranging from Data Analytics, Cloud and DevOps engineers, Computer Vision and Application development specialists.
- **Infrastructure Lead for development of "LLM as a Service" solution for Accenture customers:** Led the solution architecture, PoC and "Design for Production" of optimized infrastructure for delivering LLM as A service. This focused on NVIDIA's vGPU solution for H100, A100 GPUs on OpenShift and bare metal servers as well as Kubernetes (AKS) and RedHat OpenShift.
- **LLMaaS Platform Development Lead:** Development and delivery of a standard LLMaaS platform for government departments in Singapore, Indonesia and Thailand. I led the delivery implementation following my participation in the PoC phase and then drove the development of a "production" version of a broad range GenAI solution.
- **Developed an end-to-end Entity Extraction solution:** based on Siren and Senzing for intelligence investigation analytics. Including cloud infrastructure and automation.
- **Payments Infrastructure Modernisation Lead:** Infrastructure analysis and Modernisation Review of the NETS/FAST/GIRO/PayNow technology stack for the MAS ("Monetary Authority of Singapore") as the workstream lead for the infrastructure vertical managing two direct reports.
- **Cloud Automation Design and Implementation Lead:** Implementation of Infrastructure Code and deployment automation pipelines for managing Azure Cloud component of Hong Kong Jockey Club's Multi-Cloud build out. 6-month project. Lead SME for Azure Cloud.
- **Crypto-Currency Exchange Benchmarking Solution Development:** Implementation of a scalable performance testing and measurement solution in Go, Python, Pulsar, Kafka based on Kubernetes for a currency trading system (Eqonex, now part of Binance). 3-month project. Lead developer.

- **Cloud Governance, Security Framework and Cloud Operating Model development:** Implementation of a Cloud Governance Model and Cloud Operating model (Hong Kong Jockey Club) based on the Open Group Cloud Governance Model. Three month project. Lead architect for Cloud governance and operating model.
- **Cloud Optimisation Roadmap Development for Large Insurance Provider:** Optimisation study and recommendations for AXA Insurance Cloud Governance Framework and Cloud Operating Model including FinOps implementation. 3 month project. Lead Architect on the team managing two direct reports.

Specific Technologies used:

Technology Layer	Item 1	Item 2	Item 3	Item 4
A.I/M.L	Ollama, VLLM	LLAMA-3.3-8B, 11B, 70B, 90B, mistral, DeepSeek	Whisper (STT), Siren, Senzing	NVIDIA GPU management tools
Data	Azure file & Blob storage	Elasticsearch Vector DB	PostgreSQL	Chroma Vector DB
Observability	OpenTelemetry	OpenTrace	Prometheus	Grafana
Automation	Python-3	BASH	Golang	Terraform
Middleware	Kong A.I API Gateway	Kafka	Docker	RabbitMQ
Cloud Infrastructure	Azure AKS	Azure Cloud Networking	Azure VM + NVIDIA H100 GPUs (on Linux)	RedHat OpenShift

Senior Cloud Operations Engineer - Standard Chartered Bank

Working in an enterprise banking environment my primary responsibility was to build and operate secure, highly automated banking infrastructure in the Azure and AWS cloud.

My responsibilities were as follows:

- Detect, Report and Remediate Cloud Security Vulnerabilities discovered through vulnerability review procedures.
- Translate existing cloud architectural designs (in AWS) to Azure cloud and optimise the designs to make best use of Azure Cloud features
- Assist with the integration of FinOps Solutions and Practices (Using ApptIO Cloudability)

- Lead technical incident response for cloud service outages impacting Standard Chartered's cloud banking infrastructure
- Lead banking application deployments and upgrades for applications running in the cloud
- Develop and optimise automated processes for managing banking applications in the cloud (Operational Acceptance Testing, Security Acceptance Testing)
- Develop integration between components of legacy systems and newer cloud based systems

Additional responsibilities:

- Manage and lead a team of 4 Senior Cloud Operations Engineers/ SREs.
- Ensure Audit-Readiness of Cloud Operations Processes and lead Audit Response to MAS audits
- Operationalize Cloud Services and Solutions provided by Bank Engineering Teams
- Ensure Banking Applications onboarded to the cloud meet Bank standards for Cloud operations
- Ensure Monitoring, IAM management, Incident Response and Problem resolution processes are effective for the Cloud
- Serve as an escalation point from SREs and Cloud Support Teams in troubleshooting Cloud issues.
- Vendor Management and Cloud Governance for Azure Cloud

Senior Site Reliability Engineer - Prudential Software Services

Working in a financial services company with an enterprise environment in transition to Agile and DevOps, my detailed activities and responsibilities were as follows:

- Analyze architectural proposals by vendors for the Enterprise Hybrid Cloud implementation based on the Cloud Adoption Framework and Landing Zones
- Prepare architecture documents for Private Cloud Solutions
- Evaluate, Compare and Recommend competing vendor solutions for new systems implementation
- Develop schema and prototypes for a CMDB to support Infrastructure as Code and automation
- Write Python code to operate on a CMDB for tracking server deployments
- Write Terraform modules to build infrastructure as code in the Azure cloud
- Develop a CI/CD pipeline to automate development server deployments to cloud
- Write Ansible playbooks and roles to automate application deployment to servers
- Analyse work needed to migrate servers out of a legacy VM platform to the cloud

- Provide consulting advice to other teams on requirements for Cloud application migration
- Write technical documentation on the wiki to keep the team updated on systems changes
- Advise application teams on automating processes in the cloud using Cloud APIs and SDKs
- Troubleshoot performance issues with containerised apps running in kubernetes/OpenShift
- Root Cause and Resolve Linux server Performance Issues
- Automate Microsoft DNS management using Powershell integrating Linux systems
- Implement automated cloud backup solutions using Terraform on Azure for production systems
- Implement temporary workarounds for user automation requirements (file copies, monitoring etc)
- Perform root cause analysis on service failures, recommend solutions for failure prevention
- Define new processes for support staff to follow for handling new infrastructure support requests
- Work with Project managers as an SME to define timelines for new infrastructure projects
- Lead technical interviews for new systems engineers joining the team
- Automate kernel patches to production server clusters with zero service downtime
- Implement security compliance configuration for docker applications preparing for security audit
- Collaborate with vendors to develop Kubernetes deployment automation code for Azure
- Collaborate with vendors to implement centralized logging systems based on splunk/syslog
- Design and Propose an architecture and solution for a centralized secrets management systems based on Hashicorp Vault
- Design, Implement and maintain a total backup and restore solution for Kubernetes on Azure Cloud

Systems Engineer - Teralytics

Working in a DevOps environment with software development, infrastructure and data science teams, my responsibilities were to:

- Build and manage big data analytics clusters by gathering system requirements and carrying out the implementation and validation of the analytics stack from infrastructure to application level.

- Implementation and maintenance of Big Data processing and Analytics platforms based on Mesos, Spark, Kafka, Hadoop in AWS Cloud, on-premise virtualized, containerised and bare metal cluster environments.
- Development and support of infrastructure automation using IAC and CI/CD tools based on DevOps processes and tools (Terraform, Ansible, Jenkins and CircleCI).
- Management of Linux server clusters for streaming data ingestion, storage, processing and Analysis.
- Research and Development of solutions for improving big data processing platform efficiency, stability and usability (Spark, Hadoop and Kafka streaming, Containerization solutions).

Achievements at Teralytics:

- Completed build and operationalisation of on-premise (bare metal) analytics cluster based on apache mesos supporting spark streaming. This is used to develop and deliver analytics products for customers of Teralytics using OD matrices for rail and taxi transport route Optimisation
- Completed build and operationalisation of Virtualised/Clustered analytics cluster (VMware) based on apache mesos supporting Kafka streaming for a large ISP customer to gather retail data and deliver retail analytics.
- Operational Maintenance of Teralytics analytics clusters running on AWS cloud and bare metal systems hosted at Teralytics customers in Singapore.
- Architecture and Proposal of a replacement of Apache Mesos stack with Mesosphere's DC/OS product for tighter integration of components and smoother operation

I reported directly to the Head of Asia (Singapore) at Teralytics, with a dotted line reporting to the Data Science Lead (Singapore) and Head of Platform and Infrastructure (Zurich).

Platform Architect - noon ecommerce (under Emaar Group)

Working in a DevOps environment, my primary responsibility was to support the development of a micro service platform for ecommerce (noon.com).

Deliverables included:

- Architect, build, operate and support mesosphere DC/OS clusters on AWS which provided the containerisation platform for running around 60 micro services for ecommerce.
- Design build operate a centralised logging platform for collection and analysis of security event logs, based on Balabit syslog-ng, Splunk running on the AWS and Azure clouds as well as physical machines.

- Automate deployment and maintenance of microservice infrastructure in the AWS cloud using a Terraform, Ansible, Python (boto3), PERL and BASH, Troposphere, Cloudformation.
- Maintain primarily AWS based cloud infrastructure with some Azure components.
- Working with developers to implement continuous deployment and integration pipelines for micro services using Gitlab-CI, Jenkins, Docker and mesosphere DC/OS.
- Administer and maintain Cassandra, Kafka and Elasticsearch platforms used as the data integration core for microservices.
- Ensuring the 24x7 availability, performance and improvement of the platform,
- Implementation of new features, resolution of deployment issues, resolution of issues with micro service container integration with the platform,
- Monitoring and root cause analysis of service-impacting issues arising on the platform.

In addition, the continuity of the DC/OS platform was my responsibility in terms of re-architecture to meet cost, performance requirements and planning new container platforms based on Kubernetes, Docker swarm and AWS ECS.

My next key responsibility was to rotate between developer teams to help resolve integration issues with microservice containers and database platforms (Cassandra, PostgreSQL, MySQL, Elasticsearch) as well as messaging platforms based on Kafka.

Senior Systems Engineer (Infrastructure Services Team) - MH Alshaya

Traditional Enterprise I.T environment. Research, design, architecture and implementation of projects for delivering infrastructure services to the business, specifically:

- Scripted automation of retail data flow and processing systems using PERL/BASH/Python.
- Self-Service DBaaS solutions based on Cumulogic, Percona XtraDB, MongoDB and MySQL Fabric
- High Availability MS SQL Clustering platforms based on MSSQL Availability Groups and FCI
- Private Cloud Platforms based on Apache CloudStack 4.5, and Citrix CloudPlatform 4,5
- Highly Available, Distributed Managed File Transfer platforms for Retail information integration (based on Linux, VSFTPD, Proftpd and JSCAPE)
- Distributed Vulnerability Scanning Clusters based on OpenVAS 8
- Hadoop enterprise data hub (MAPR, Cloudera) to support Alshaya's Big Data R&D needs
- Poc and Investigate PaaS solutions based on CloudFoundry and OpenShift
- VMware and HyperV Automation using custom scripting and VM templating.
- Centralized Authentication for integrating linux and solaris systems with Active Directory

- Configuration management and patch deployment system for linux and solaris servers (Ansible and BASH)
- Centralized Logging systems based on Splunk and Syslog-ng, Elasticsearch and Kibana.
- Development and Support of secure Point of Sale operating systems based on Linux for deployment in Alshaya's stores.

Operations Engineer (EC2 Platform) - Amazon Web Services

- Analysis of monitoring data across the global EC2 cloud to identify emerging problem trends and feed into continuous improvement cycle.
- Detailed root cause analysis of failures on the EC2 platform.
- Driving EC2 platform issues to resolution by gathering data and coordinating solution development with responsible development teams
- Reviewing change requests for maintenance across the global AWS EC2 cloud.
- Assisting resolution of large scale maintenance on AWS EC2 virtual server platform
- Reporting on key metrics for the global EC2 platform in monthly metrics reviews.
- Resolving enterprise customer queries escalated from the AWS Cloud Support Division

Cloud Support Engineer - Amazon Web Services

A member of the AWS Cloud Support team based in Cape Town I worked for the world leader in cloud computing, helping companies take their computing infrastructure and applications into the AWS cloud.

My daily work consisted of:

- Learning and using ground-breaking Cloud and Virtualization technologies, both proprietary (AWS's SaaS, IaaS and PaaS products) and Open Source to provide cloud solutions to AWS' customers.
- Cloud solution architecture (Implementing and design of AWS cloud based solutions)
- Cloud Support and System Integration using the AWS product line
- Weekly interviews with potential candidates for engineering positions at AWS
- Assisting core AWS engineering teams to assist in the support and maintenance of the AWS EC2 cloud infrastructure itself
- Testing, developing and using server fleet management tools to maintain thousands of customer Cloud Server instances on a massive scale.

UNIX Systems Engineer (Infrastructure Services) - MTN Business Solutions Pty. Ltd

My responsibilities and contributions during my time at MTN were:

- Fault-finding, setup, configuration, maintenance and administration of MTN Business UNIX (FreeBSD, Solaris, Debian and RHEL) Servers in countrywide datacenters.
- Upgrade the MTN Business service servers keeping up with the international trends specifically with regards to DNS, SMTP, RADIUS and other infrastructure services
- Administer, Maintain and Improve HP StorageWorks, Hitachi Datadomain appliances, Dell MD3000 and Hitachi Adaptive Storage based Virtualized Storage systems.
- Maintain, Administer and develop monitoring/management tools for VMware ESXi Virtualisation clusters
- Design, Implement and Maintain Server, Application and Service monitoring systems integrated with Netcool, Nagios and Net-SNMP for capacity management and service monitoring.
- Provide general UNIX Systems Administration services to all other teams within the Information Systems division in order to integrate software systems and processes.

Network Monitoring Specialist - TENET

Working with a small core team of network engineers, my role was to implement and operate monitoring for a countrywide DWDM network serving South African Universities and research institutions. My responsibilities were to:

- Develop integrated NMS solutions for fault and performance monitoring of the country wide and international WAN using Open source Software on Linux/ Solaris platforms, with PERL, ruby and python.
- Develop custom routing performance monitoring systems (for BGP) and network availability reporting tools for use by Service Management and NOC staff.
- Develop and document an ITIL compliant CMDB (configuration management database) to support the ISP operations of TENET, using PERL to integrate various database-backed systems, and a combination of PHP and RoR to implement user interfaces.
- Provide scalable automated solutions, using PERL, for the administration of UNIX/Linux servers used to manage the TENET network.
- Develop web-based Change Control Systems in PERL and PHP. Provide integration with Service Desk/Help Desk Systems.
- Carry out the function of Acting Service Manager for the TENET network in the absence of the Service Manager.

3. Education and Qualifications

Formal Education

Educational Level	Institution	Start Date	End date	Duration	Status
Matriculation	Haythorne High School (South Africa)	1991	1996	5 years	Complete
Tertiary/University: Bsc. Electrical Engineering (Light Current)	University of Cape Town (South Africa)	1997	2000	4 years	INCOMPLETE

Employer Provided Training Programmes:

Item	Course/Certification	Institution
1	ITIL v3 Foundations Certification	HP Education Services
2	Red Hat Enterprise Linux Training Fast-Track	Sybase Training Services
3	LPIC L1 Certification	Pearson Prometric/ Linux professional institute
4	Perl Programming	HP Education Services
5	Advanced DNS Infrastructure Training/DNSSEC Infrastructure	UNIFORUM S.A
6	Cisco UCS Administration (Training only)	Cisco

4. Technical Proficiency Matrix

The following table summarises the range of specific technologies I've worked with.

General Technology Areas

Area	Experience (years)	Technology Platforms	Last practiced
System Benchmarking, Performance analysis and Tuning	15	Azure Cloud, Kubernetes, Docker, Spark, Linux, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, VMware, Hyper-V	Current

High Availability Solution Design and Architecture	18	Azure Cloud, Linux, Kubernetes, Docker Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, Hyper-V, Spark	Current
Automation Scripting for System Administration and Configuration	18	Azure Cloud, Linux, Kubernetes, Spark, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, VMware, Hyper-V	Current
O.S, Network and Server Troubleshooting and Analysis	18	Azure Cloud, Linux, Kubernetes, Spark, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, VMware, Hyper-V	Current
Network and System Monitoring implementation and management	18	Azure Cloud, Linux, Kubernetes, Spark, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, VMware, Hyper-V	Current
Technical Documentation	18	JIRA, Confluence, Git	Current
Server and Network Security Implementation	18	Azure Cloud, Linux, Kubernetes, Spark, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, Rackspace Cloud, VMware, Hyper-V	Current
I.T Solution Support and Consulting	18	Azure Cloud, Linux, Docker, Kubernetes, Spark, Mesos, HDFS, CephFS, Solaris, FreeBSD, AWS Cloud Solutions	Current
Technical Root Cause Analysis	16	Azure Cloud, Linux, Solaris, FreeBSD, Mac OSX, AWS Cloud Solutions, VMware, Hyper-V	Current

6. Certifications

- AI-102: Designing and Implementing Microsoft AI Solutions
- AI-900: Azure-Artificial Intelligence Fundamentals Certification
- NVIDIA AI Infrastructure and Operations Fundamentals
- Intel MLOps Certified Professional
- Site Reliability Engineering (SRE) Foundation v1.1
- CKAD: Certified Kubernetes Application Developer
- AZ-304 Microsoft Certified: Azure Solutions Architect Expert
- AZ-303 Microsoft Azure Architect Technologies
- FOCF: FinOps Certified Practitioner
- LPIC-1: LPI Level 1 Linux Administration
- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: Azure Developer Associate
- HashiCorp Certified: Consul Associate
- HashiCorp Certified: Vault Associate
- HashiCorp Certified: Terraform Associate
- ITIL v3 Foundation
- SC-900: Azure Security Fundamentals
- PCAP I: /Certified Associate in Python Programming

7. Conferences

- Nvidia GTC 2026 (represented Firmus at Firmus Booth and represented the CTO at the Nvidia CTO round-table hosted by Nvidia's CTO)